NOTRE DAME UNIVERSITY
BANGLADESH

Computer Networks Hardware Lab Report

Course Code: CSE-3204
Course Title: Computer Networks Lab
Lab Task Topic: Introduction to Hardware Equipments

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Submission Date: April 24, 2025

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Introduction

The rapid growth of computer networks and internet-based technologies has significantly increased the demand for reliable and high-performance network infrastructure. Understanding the physical hardware components that form the foundation of such networks is essential for any networking professional or student. This lab report provides a comprehensive overview of various fundamental hardware tools and equipment used in networking environments.

In this report, we explore the practical application and functionality of essential devices such as cable tracers, cable testers, RS-232 cables, fiber optic duplex cables, and SFP modules. We also study network components like switches and routers, along with their configurations and interconnections. A key focus is placed on the T-568B Ethernet cabling standard, which defines the proper arrangement of wire pairs in twisted-pair cables for effective network communication.

Additionally, the report covers the Open Systems Interconnection (OSI) model, a conceptual framework that categorizes and standardizes the functions of a communication system into seven distinct layers. Each hardware topic is supported with descriptions and relevant images to aid in practical understanding.

This lab aims to enhance students' hands-on skills in identifying, using, and configuring networking hardware, thereby building a strong foundation for designing and managing robust network systems.

1 Cable Tracer

A **Cable Tracer** is used to trace and identify cables or wires within a group without damaging the insulation. It is useful for locating breaks or shorts in cables and identifying signal paths.

2 Cable Tester

A **Cable Tester** is a device used to test the strength and connectivity of a particular type of cable or other wired assemblies. It checks for continuity, wiring configuration, and signal loss.

3 T-568B Wiring Standard

The **T-568B** standard is a wiring scheme used for terminating network cables (Ethernet) using RJ-45 connectors. The order of colored wires is:

1. White Orange
2. Orange

3. White Green
4. Blue
5. White Blue
6. Green
7. White Brown
8. Brown

4 RS-232 Cable

The **RS-232 Cable** is used for serial communication between devices. It was commonly used for connecting PCs to modems and is still used in networking hardware for console connections.

5 Fiber Optic (Duplex) Cable

A **Fiber Optic Duplex Cable** consists of two fiber optic strands in a single cable, used for bidirectional data transfer. It allows high-speed and long-distance data communication.

6 SFP Module

The **Small Form-factor Pluggable (SFP)** module is a hot-swappable transceiver used in networking devices for fiber optic and Ethernet connections.

7 Switch

A **Switch** is a network device that connects devices in a LAN and uses MAC addresses to forward data to the correct destination.

8 Router Configuration

A **Router** connects different networks and directs data between them. Configuration typically involves setting IP addresses, routing protocols, and NAT.

Example CLI Commands:

```
interface fastethernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

9 Switch to Router Connection

Switch to Router Connection is usually done through a straight-through Ethernet cable. The router's LAN port connects to the switch, enabling device communication and internet access.

10 OSI Model

The **OSI (Open Systems Interconnection) Model** is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven abstraction layers:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application